

Solving Artuc, Chaudhuri, and McLaren 2010 model

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See Artuc and Ortega 2026 for a detailed discussion on solving trade models using contraction mapping property of the equilibrium system. In this note, I am considering the simplest case where all sectors are traded and prices are exogenous.

Here are the steps:

- **Step 0:** First, guess V_t^i and L_t^i for $i \in \{1, 2, \dots, J\}$ and $t \in \{1, 2, \dots, T\}$.
- **Step 1:** Calculate the consumer price index

$$\phi_t = \prod (p_t^i)^{\theta^i},$$

and calculate real wages from MPL given labor allocation

$$w_t^i = \frac{p_t^i}{\phi_t} \alpha^i \psi^i \left((L_t^i)^{\rho^i - 1} \right) \left(\alpha^i (L_t^i)^{\rho^i - 1} + (1 - \alpha^i) \right)^{\frac{1}{\rho^i} - 1},$$

from Equation 14 in the paper.

- **Step 2:** Calculate moving probabilities

$$m_t^{ij} = \frac{\exp \left([\beta V_t^j - C^{ij}] / \nu \right)}{\sum_k \exp \left([\beta V_t^k - C^{ik}] / \nu \right)},$$

from Equation 10 in the paper.

- **Step 3:** Calculate option values

$$\Omega_t^i = -\nu \log (m_t^{ii}),$$

from Equation 7.

- **Step 4:** Implied values

$$V_t^{i*} = w_t^i + \beta V_{t+1}^i + \Omega_t^i,$$

from Equation 3.

- **Step 5:** Implied labor allocations

$$L_{t+1}^{i*} = \sum_k L_t^k m_t^{ki},$$

(we did not show this equation in the paper I think, but it is simply the labor flows).

- **Step 6:** Calculate errors

$$\begin{aligned}\xi_1 &= \sum_i \sum_t |V_t^{i*} - V_t^i|, \\ \xi_2 &= \sum_i \sum_t |L_t^{i*} - L_t^i|.\end{aligned}$$

Update guessed arrays

$$\begin{aligned}V_t^i &= \vartheta V_t^{i*} + (1 - \vartheta)V_t^i, \\ L_t^i &= \vartheta L_t^{i*} + (1 - \vartheta)L_t^i,\end{aligned}$$

where $\vartheta = 0.4$ is the update speed. We are working with real variables only, so no need to scale or normalize. If both $\xi_1 < 10^{-8}$ and $\xi_2 < 10^{-8}$ then stop, else go to Step 1.

References

- Artuc, Erhan, Shubham Chaudhuri, and John McLaren (2010). “Trade shocks and labor adjustment: A structural empirical approach”. In: *American Economic Review* 100.3, pp. 1008–45.
- Artuc, Erhan and Johan Ortega (2026). “International Trade Policy and Quantitative Models: A Practitioner’s Guide”. In: *World Bank Policy Research Working Paper Series*.